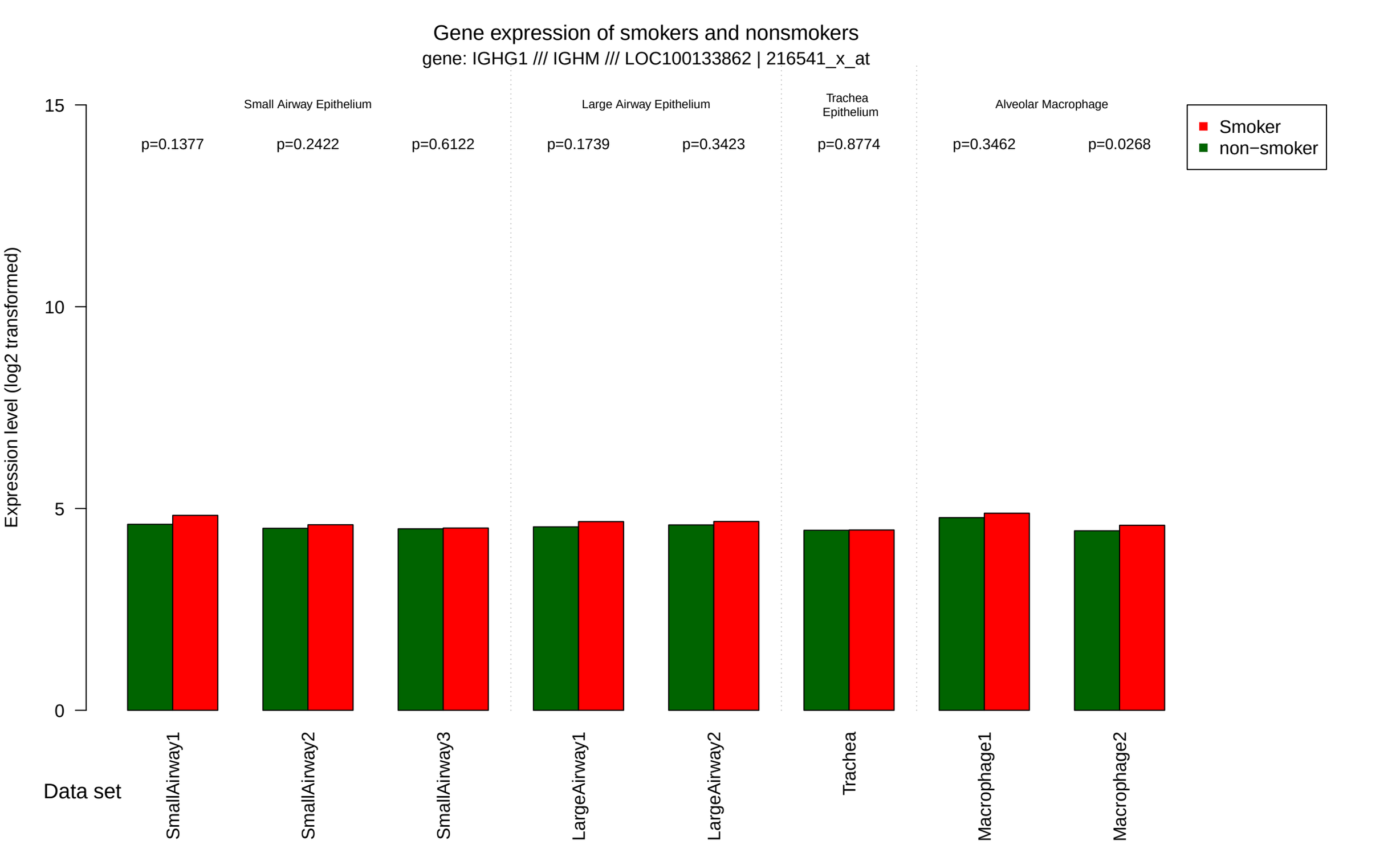
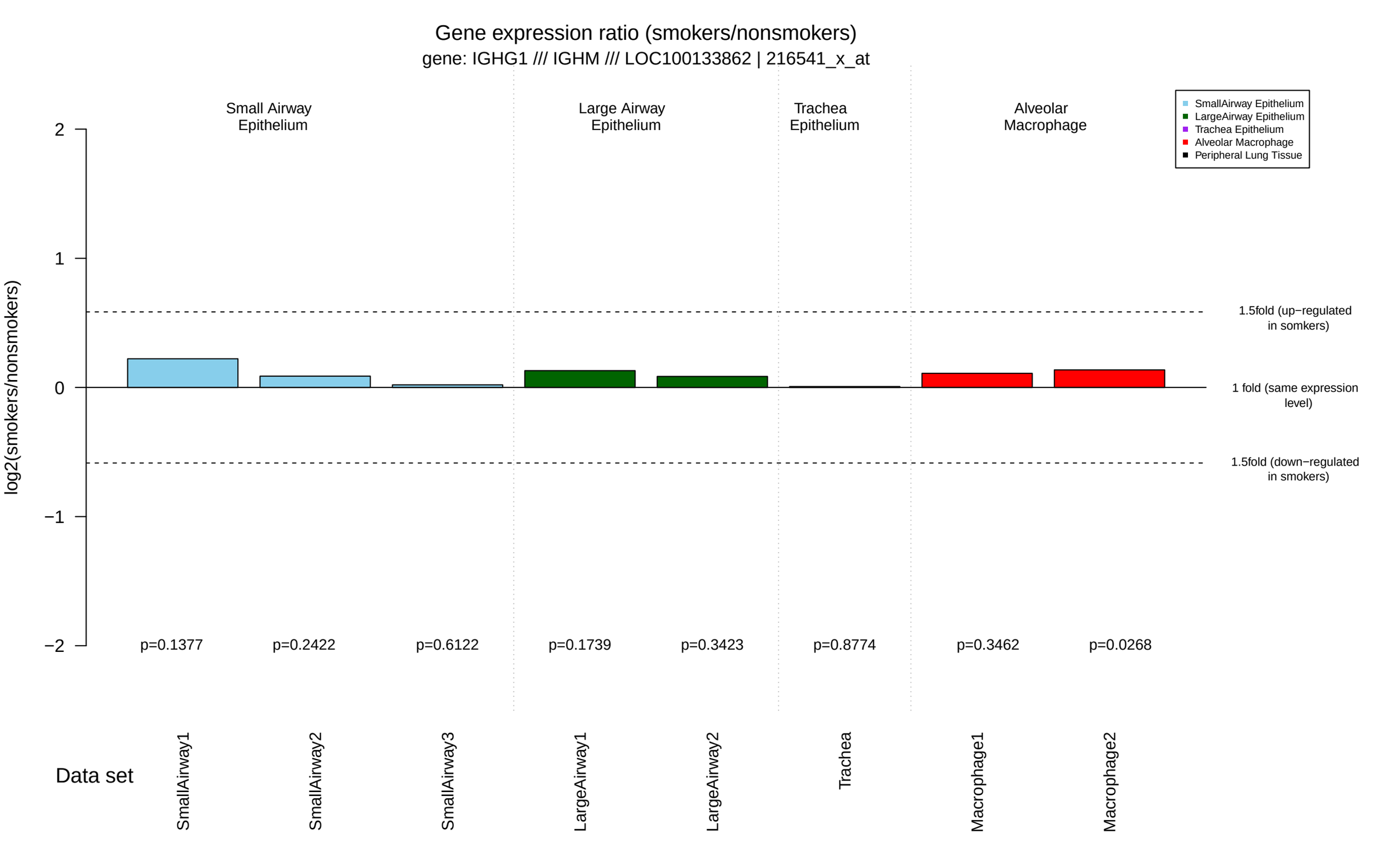
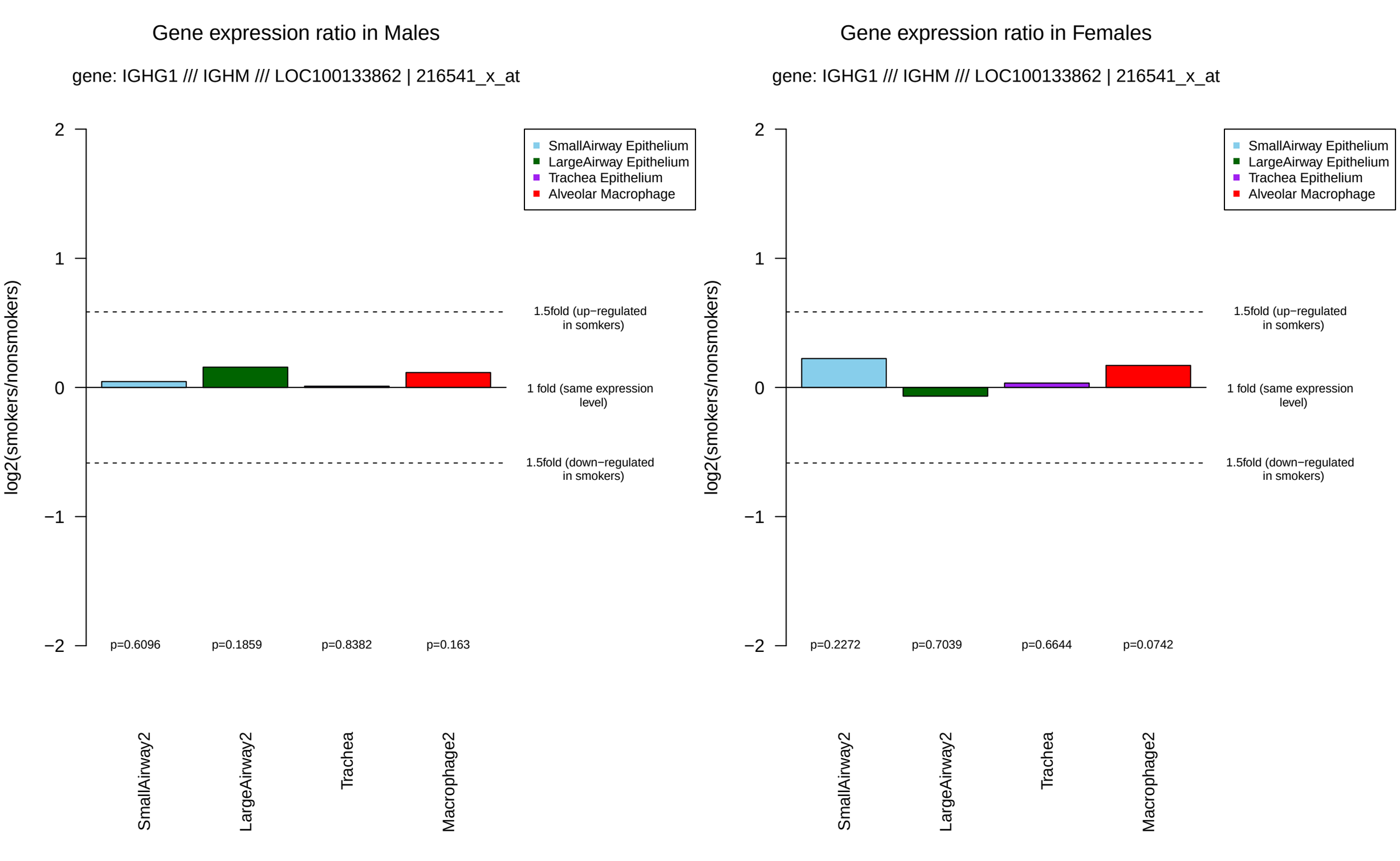
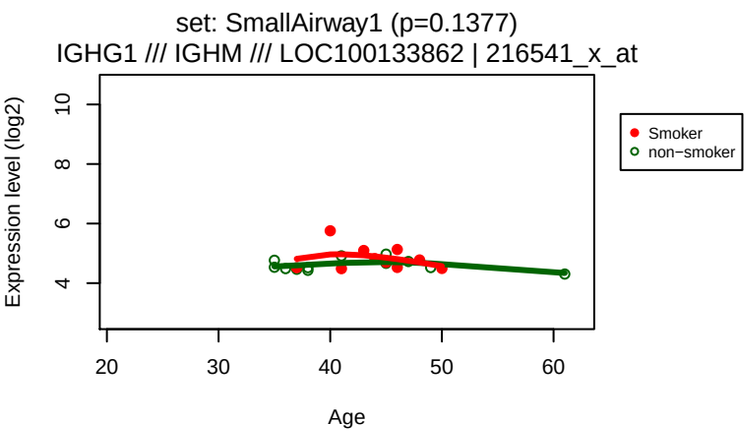
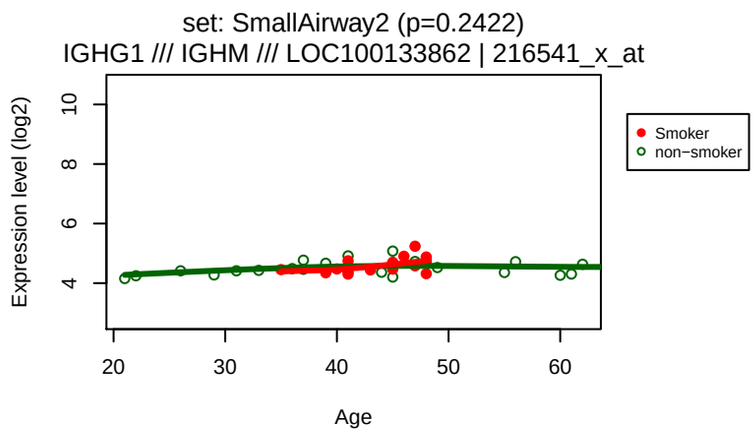
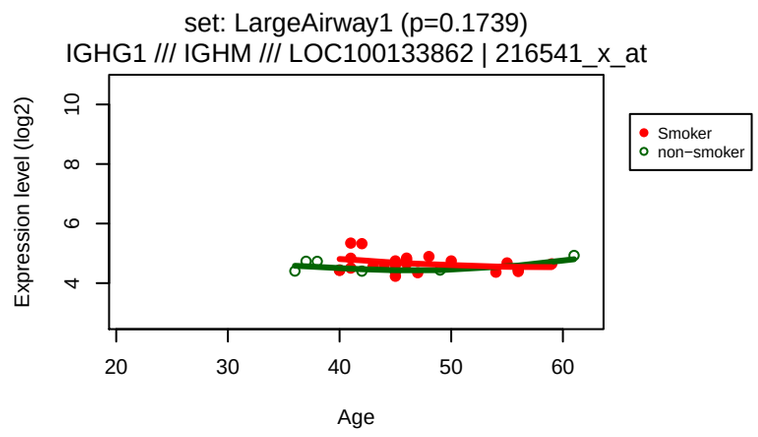

Gene expression of smokers and nonsmokers
gene: IGHG1 /// IGHM /// LOC100133862 | 216541_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1377
p=0.2422
p=0.6122
p=0.1739
p=0.3423
p=0.8774
p=0.3462
p=0.0268
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

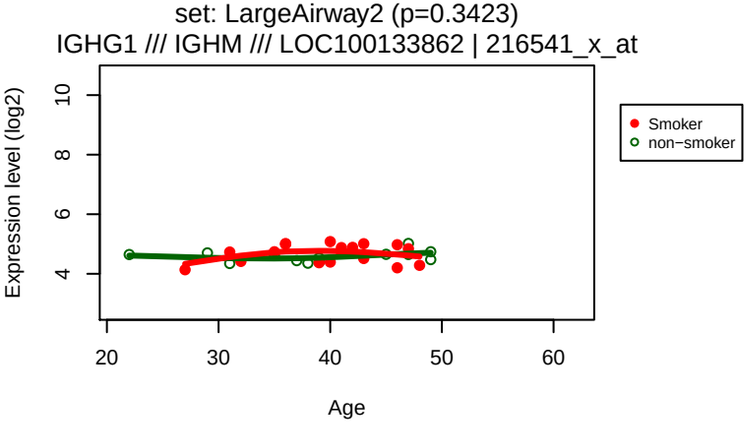

Gene expression ratio (smokers/nonsmokers)
gene: IGHG1 /// IGHM /// LOC100133862 | 216541_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1377
p=0.2422
p=0.6122
p=0.1739
p=0.3423
p=0.8774
p=0.3462
p=0.0268
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

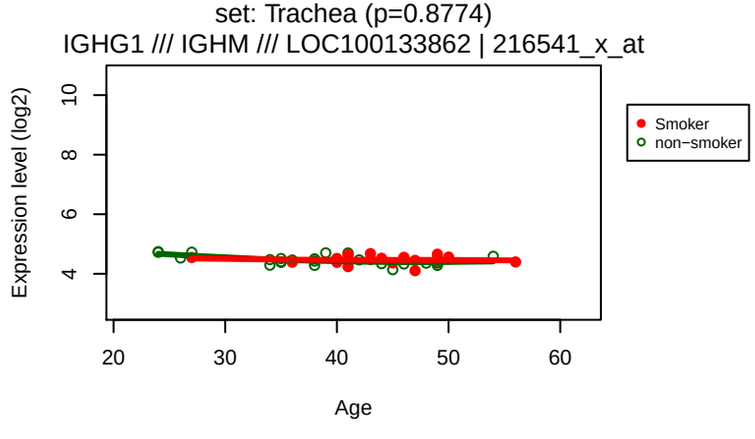

Gene expression ratio in Males
gene: IGHG1 /// IGHM /// LOC100133862 | 216541_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6096
p=0.1859
p=0.8382
p=0.163
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IGHG1 /// IGHM /// LOC100133862 | 216541_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2272
p=0.7039
p=0.6644
p=0.0742
SmallAirway2
LargeAirway2
Trachea
Macrophage2

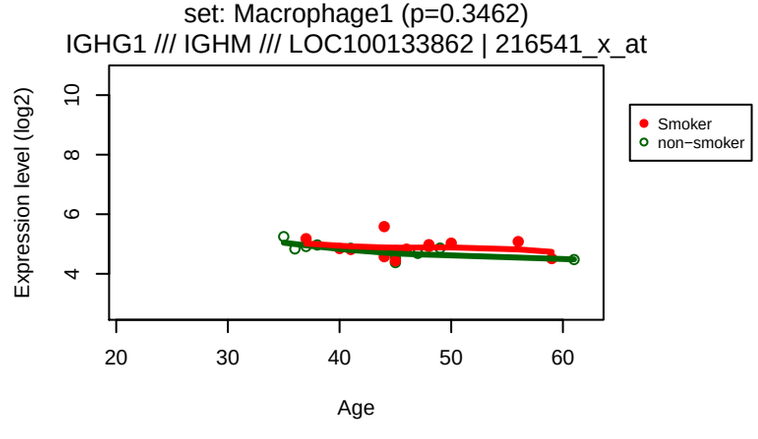

set: SmallAirway1 (p=0.1377)
IGHG1 /// IGHM /// LOC100133862 | 216541_x_at
Expression level (log2)
Age
Smoker
non−smoker

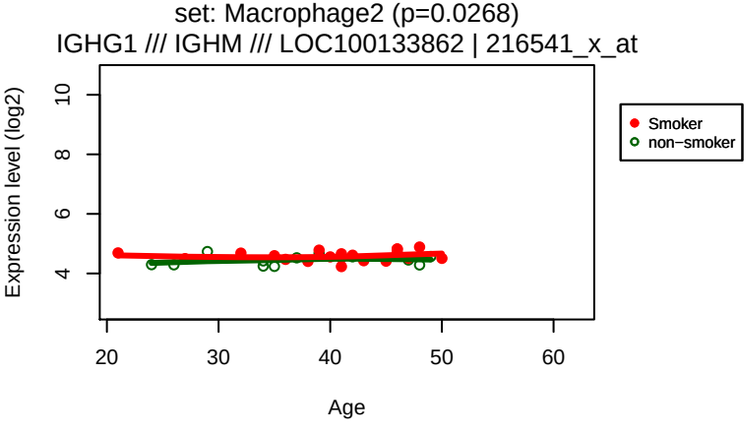

set: SmallAirway2 (p=0.2422)
IGHG1 /// IGHM /// LOC100133862 | 216541_x_at
Expression level (log2)
Age
Smoker
non−smoker

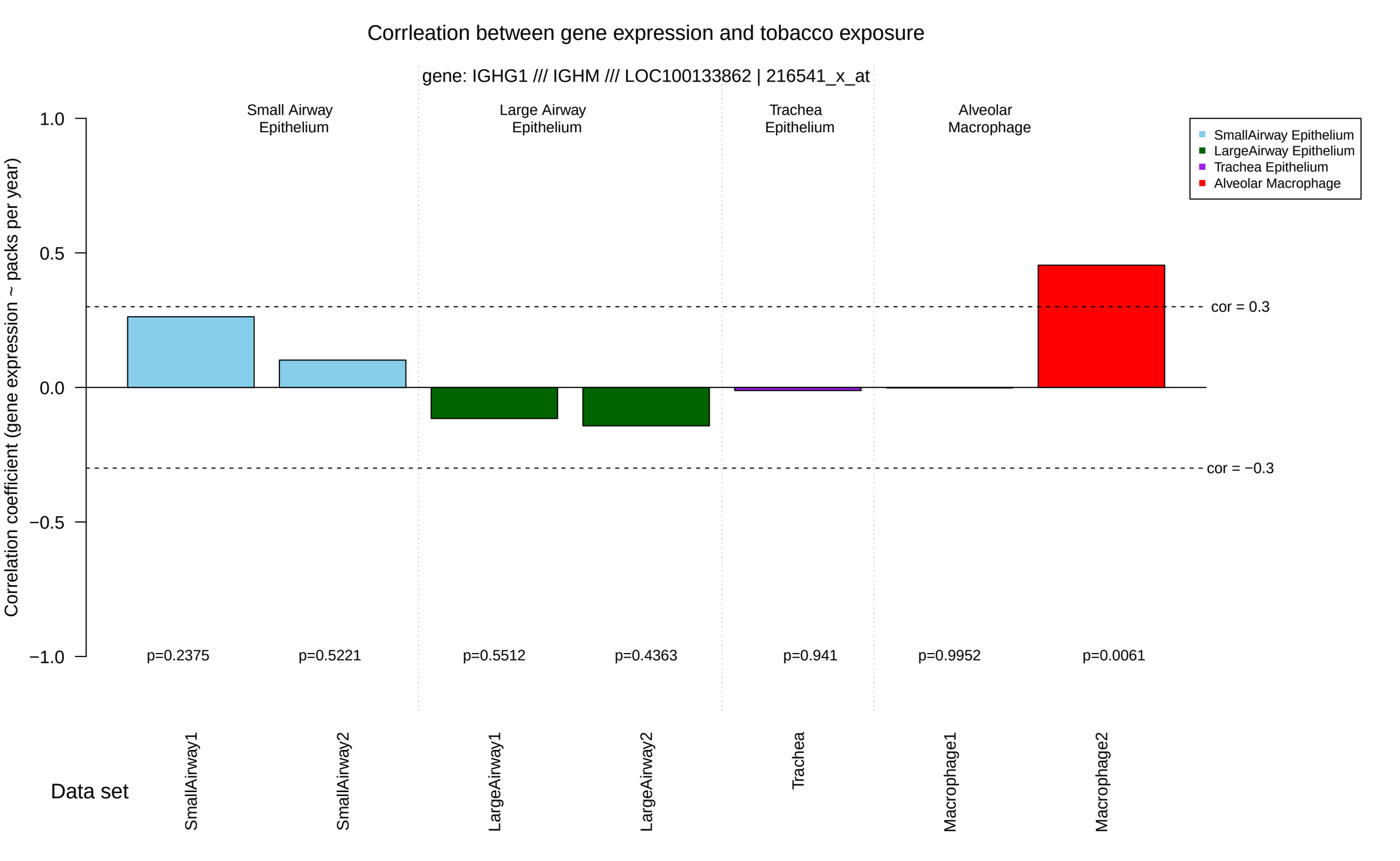

set: LargeAirway1 (p=0.1739)
IGHG1 /// IGHM /// LOC100133862 | 216541_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3423)
IGHG1 /// IGHM /// LOC100133862 | 216541_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8774)
IGHG1 /// IGHM /// LOC100133862 | 216541_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3462)
IGHG1 /// IGHM /// LOC100133862 | 216541_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0268)
IGHG1 /// IGHM /// LOC100133862 | 216541_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IGHG1 /// IGHM /// LOC100133862 | 216541_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
Correlation coefficient (gene expression ~ packs per year)
cor = 0.3
cor = −0.3
p=0.2375
p=0.5221
p=0.5512
p=0.4363
p=0.941
p=0.9952
p=0.0061
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2